

Route Optimization Platform

Team 9 — Project Presentation
Assignment 4



Project context

Problem

Logistics companies need to plan vehicle and loader routes for delivery scenarios with time windows, capacity constraints, and optional orders.

Solution

A web-based route optimization platform: submit a JSON problem instance, get optimized vehicle and loader routes — with automated validation, job history, and a visual interface.

Stakeholders

- Dispatcher / logistics manager — primary end-user
- Client — drivers and loaders
- Course instructors — evaluators



Demo

Route Optimizer Dashboard New Job Validate API Docs

Jobs + New Job

ID	NAME	STATUS	CREATED	OBJECTIVE	ACTIONS
2428684c...	—	Completed	27.06.2026, 12:40:31	98084.48000000001	View
5f7f5684...	—	Completed	27.06.2026, 08:43:44	30013.5156	View
a891c610...	—	Completed	21.06.2026, 21:40:59	29683.8646	View
c15bc6b9...	—	Completed	21.06.2026, 21:37:12	29683.8646	View
18d8453b...	123	Completed	21.06.2026, 21:24:37	28647.8896	View
84af21e5...	123	Completed	21.06.2026, 21:22:55	28647.8896	View
b774803d...	—	Completed	21.06.2026, 21:16:51	28647.8896	View
56558b88...	—	Completed	21.06.2026, 21:15:58	29683.8646	View
eaef51d3b...	123	Completed	21.06.2026, 21:13:28	28647.8896	View
7c3b4a24...	123	Completed	21.06.2026, 21:09:51	28647.8896	View
918892db...	—	Completed	21.06.2026, 18:34:33	30013.5156	View

https://drive.google.com/file/d/15Dh_azNvTxptEjW1XX4S_jnOmg84rHg/view?usp=sharing



Team

Name	Role	Contribution
Elvina Belorusova	Scrum Master	Documentation, Organisation of work
Adeliia Verenikina	Product Owner	Documentation, Repo workflow, Presentation
Aydar Gaifullin	Developer	Backend, Solver, Algorithm
Matvei Trifonov	Developer	Backend, Solver, VM setup
Valerii Verbovetc	Developer	DevOps, Deployment, API, Integration, Frontend



Requirements

Must Have — Done

- Vehicle and loader route output
- Hard constraint validation
- Docker execution (Docker Compose: api + frontend)
- REST API (FastAPI + OR-Tools)

Should Have — Done

- Algorithm time limit and reproducible seed
- Objective function value
- React web frontend

Could Have — In Progress

- Skipped optional orders report
- Route visualization improvements

ID	Requirement	Threshold
QRT-FC-01	Solver functional correctness	0 violations
QRT-PE-01	Solver time behaviour	≤ 900 seconds
QRT-RE-01	Save jobs in database	No data loss on restart
QRT-SE-01	Safe error confidentiality	No tracebacks or paths in API response



Roadmap

Done — MVP v0

REST API, OR-Tools solver pipeline, Docker

Done — MVP v1

React SPA, route visualization, validation endpoint, health endpoint, job listing, Docker Compose

Current — Assignment 4

Persistent storage, quality requirement tests, CI quality gates, solver robustness, new SemVer release

Next Sprint

Solver scalability on large instances, job retention policies, extended frontend testing, UAT feedback fixes

Future

Authentication, multi-tenancy, async worker queue, production monitoring



Client Collaboration

Validated by client

- OR-Tools solver approach reviewed and positively received by the client
- REST API and React UI reviewed and accepted as MVP v1

Useful feedback received

- Optional-order selection should account for loader costs earlier in the algorithm
- Solution quality on small instances needs improvement
- Coordinate-plane visualization is appropriate (benchmark data has no real geographic coordinates)
- Persistent calculation history is desired

Current challenges

- Solver quality on small instances
- Persistent storage not yet reflected in Docker Compose
- Skipped optional orders report deferred to Sprint 3



Links & QR Codes



<https://github.com/Team-9-swp/Route-optimization-Platform-swp>



<http://10.93.26.188:3000/>

